

Compendium of International Methods of Wine and Must Analysis

Download document

OIVStandards and technical documentsCompendium of International Methods of Wine and Must AnalysisAnnex A - Methods of Analysis of Wines and MustsSECTION 3 - CHEMICAL ANALYSIS

SECTION 3.1 - ORGANIC COMPOUNDSSECTION 3.1.2 - ALCOHOLS

Tables of correction

OIV-MA-AS312-02 Alcoholic strength by volume

| 1 International alcoholic strength at 20°C<br>of apparent densities of ethanol-water mixtures - <u>Pyrex pycnometer</u> Densities at t°C, corrected for air buoyancy |        |        |        |        |        |        |        |        |        |        |        |        |        |        |        |        |        |        |        |        |        |        |        |      |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|------|
| Alcohol % by volume                                                                                                                                                  |        |        |        |        |        |        |        |        |        |        |        |        |        |        |        |        |        |        |        |        |        |        |        |      |
| 0                                                                                                                                                                    |        | 1      |        | 2      |        | 3      |        | 4      |        | 5      |        | 6      |        | 7      |        | 8      |        | 9      |        | 10     |        | 11     |        |      |
| 999.64                                                                                                                                                               | 1.50   | 998.14 | 1.44   | 996.70 | 1.40   | 995.30 | 1.35   | 993.95 | 1.30   | 992.65 | 1.24   | 991.41 | 1.19   | 990.22 | 1.14   | 989.08 | 1.10   | 987.98 | 1.05   | 986.93 | 1.00   | 985.93 | 0.95   |      |
| -0.07                                                                                                                                                                |        | -0.06  |        | -0.06  |        | -0.06  |        | -0.06  |        | -0.06  |        | -0.06  |        | -0.05  |        | -0.04  |        | -0.03  |        | -0.02  |        | -0.01  |        |      |
| 999.71                                                                                                                                                               | 1.51   | 998.20 | 1.44   | 996.76 | 1.40   | 995.36 | 1.35   | 994.01 | 1.30   | 992.71 | 1.24   | 991.47 | 1.20   | 990.27 | 1.15   | 989.12 | 1.11   | 988.01 | 1.06   | 986.95 | 1.01   | 985.94 | 0.97   |      |
| -0.05                                                                                                                                                                |        | -0.05  |        | -0.04  |        | -0.04  |        | -0.04  |        | -0.04  |        | -0.03  |        | -0.03  |        | -0.02  |        | -0.02  |        | -0.01  |        | 0.00   |        |      |
| 999.76                                                                                                                                                               | 1.51   | 998.25 | 1.45   | 996.80 | 1.40   | 995.40 | 1.35   | 994.05 | 1.30   | 992.75 | 1.25   | 991.50 | 1.20   | 990.30 | 1.16   | 989.14 | 1.11   | 988.03 | 1.07   | 986.96 | 1.02   | 985.94 | 0.98   |      |
| -0.03                                                                                                                                                                |        | -0.03  |        | 4.03   |        | -0.02  |        | -0.02  |        | -0.02  |        | -0.02  |        | -0.01  |        | -0.01  |        | 0.00   |        | 0.01   |        | 0.02   |        |      |
| 3°                                                                                                                                                                   | 999.79 | 1.51   | 998.28 | 1.45   | 996.83 | 1.41   | 995.42 | 1.35   | 994.07 | 1.30   | 992.77 | 1.25   | 991.52 | 1.21   | 990.31 | 1.16   | 989.15 | 1.12   | 988.03 | 1.08   | 986.95 | 1.03   | 985.92 | 1.00 |
| -0.02                                                                                                                                                                |        | -0.02  |        | -0.01  |        | -0.02  |        | -0.01  |        | -0.01  |        | 0.00   |        | 0.00   |        | 0.01   |        | 0.02   |        | 0.03   |        | 0.04   |        |      |
| 4°                                                                                                                                                                   | 999.81 | 1.51   | 998.30 | 1.46   | 996.84 | 1.40   | 995.44 | 1.36   | 994.08 | 1.30   | 992.78 | 1.26   | 991.52 | 1.21   | 990.31 | 1.17   | 989.14 | 1.13   | 988.01 | 1.09   | 986.92 | 1.04   | 985.88 | 1.00 |
| 0.00                                                                                                                                                                 |        | 0.00   |        | 0.00   |        | 0.00   |        | 0.01   |        | 0.02   |        | 0.02   |        | 0.02   |        | 0.02   |        | 0.03   |        | 0.04   |        | 0.05   |        |      |
| 5°                                                                                                                                                                   | 999.81 | 1.51   | 998.30 | 1.46   | 996.84 | 1.40   | 995.44 | 1.37   | 994.07 | 1.31   | 992.76 | 1.26   | 991.50 | 1.21   | 990.29 | 1.17   | 989.12 | 1.14   | 987.98 | 1.10   | 986.88 | 1.05   | 985.83 | 1.01 |
| 0.01                                                                                                                                                                 |        | 0.01   |        | 0.01   |        | 0.02   |        | 0.01   |        | 0.02   |        | 0.03   |        | 0.04   |        | 0.05   |        | 0.05   |        | 0.05   |        | 0.06   |        |      |
| 6°                                                                                                                                                                   | 999.80 | 1.51   | 998.29 | 1.46   | 996.83 | 1.41   | 995.42 | 1.36   | 994.06 | 1.32   | 992.74 | 1.27   | 991.47 | 1.22   | 990.25 | 1.18   | 989.07 | 1.14   | 987.93 | 1.10   | 986.83 | 1.06   | 985.77 | 1.03 |
| 0.03                                                                                                                                                                 |        | 0.03   |        | 0.03   |        | 0.03   |        | 0.04   |        | 0.04   |        | 0.04   |        | 0.05   |        | 0.06   |        | 0.07   |        | 0.08   |        | 0.09   |        |      |
| 7°                                                                                                                                                                   | 999.77 | 1.51   | 998.26 | 1.46   | 996.80 | 1.41   | 995.39 | 1.37   | 994.02 | 1.32   | 992.70 | 1.27   | 991.43 | 1.23   | 990.20 | 1.19   | 989.01 | 1.15   | 987.86 | 1.11   | 986.75 | 1.07   | 985.68 | 1.03 |
| 0.05                                                                                                                                                                 |        | 0.04   |        | 0.04   |        | 0.05   |        | 0.05   |        | 0.05   |        | 0.05   |        | 0.06   |        | 0.06   |        | 0.07   |        | 0.08   |        | 0.09   |        |      |
| 8°                                                                                                                                                                   | 999.72 | 1.50   | 998.22 | 1.46   | 996.76 | 1.42   | 995.34 | 1.37   | 993.97 | 1.32   | 992.65 | 1.27   | 991.38 | 1.24   | 990.14 | 1.19   | 988.95 | 1.16   | 987.79 | 1.12   | 986.67 | 1.08   | 985.59 | 1.05 |
| 0.05                                                                                                                                                                 |        | 0.06   |        | 0.06   |        | 0.06   |        | 0.06   |        | 0.06   |        | 0.07   |        | 0.07   |        | 0.08   |        | 0.09   |        | 0.10   |        | 0.11   |        |      |
| 9°                                                                                                                                                                   | 999.67 | 1.51   | 998.16 | 1.46   | 996.70 | 1.42   | 995.28 | 1.37   | 993.91 | 1.32   | 992.59 | 1.28   | 991.31 | 1.24   | 990.07 | 1.20   | 988.87 | 1.17   | 987.70 | 1.13   | 986.57 | 1.09   | 985.48 | 1.06 |
| 0.07                                                                                                                                                                 |        | 0.07   |        | 0.07   |        | 0.07   |        | 0.07   |        | 0.08   |        | 0.08   |        | 0.09   |        | 0.09   |        | 0.10   |        | 0.11   |        | 0.12   |        |      |
| 10°                                                                                                                                                                  | 999.60 | 1.51   | 998.09 | 1.46   | 996.63 | 1.42   | 995.21 | 1.37   | 993.84 | 1.33   | 992.51 | 1.28   | 991.23 | 1.25   | 989.98 | 1.20   | 988.78 | 1.17   | 987.60 | 1.14   | 986.46 | 1.10   | 985.36 | 1.06 |
| 0.09                                                                                                                                                                 |        | 0.09   |        | 0.09   |        | 0.08   |        | 0.09   |        | 0.09   |        | 0.10   |        | 0.10   |        | 0.11   |        | 0.11   |        | 0.12   |        | 0.13   |        |      |
| 11°                                                                                                                                                                  | 999.51 | 1.51   | 998.00 | 1.46   | 996.54 | 1.41   | 995.13 | 1.38   | 993.75 | 1.33   | 992.42 | 1.29   | 991.13 | 1.25   | 989.88 | 1.21   | 988.67 | 1.18   | 987.49 | 1.15   | 986.34 | 1.11   | 985.23 | 1.07 |

|     |        |      |        |      |        |      |        |      |        |      |        |      |        |      |        |      |        |      |        |      |        |      |        |      |
|-----|--------|------|--------|------|--------|------|--------|------|--------|------|--------|------|--------|------|--------|------|--------|------|--------|------|--------|------|--------|------|
| 12° | 999.41 | 1.50 | 997.91 | 1.46 | 996.45 | 1.42 | 995.03 | 1.38 | 993.65 | 1.34 | 992.31 | 1.29 | 991.02 | 1.25 | 989.77 | 1.22 | 988.55 | 1.19 | 987.36 | 1.15 | 986.21 | 1.12 | 985.09 | 1.09 |
|     | 0.11   |      | 0.11   |      | 0.11   |      | 0.11   |      | 0.11   |      | 0.11   |      | 0.12   |      | 0.12   |      | 0.13   |      | 0.14   |      | 0.15   |      | 0.16   |      |
| 13° | 999.30 | 1.50 | 997.80 | 1.46 | 996.34 | 1.42 | 994.92 | 1.38 | 993.54 | 1.34 | 992.20 | 1.30 | 990.90 | 1.25 | 989.65 | 1.23 | 988.42 | 1.20 | 987.22 | 1.16 | 986.06 | 1.13 | 984.93 | 1.09 |
|     | 0.12   |      | 0.12   |      | 0.13   |      | 0.13   |      | 0.13   |      | 0.13   |      | 0.14   |      | 0.14   |      | 0.15   |      | 0.16   |      | 0.16   |      | 0.16   |      |
| 14° | 999.18 | 1.50 | 997.68 | 1.46 | 996.22 | 1.43 | 994.79 | 1.38 | 993.41 | 1.34 | 992.07 | 1.30 | 990.77 | 1.26 | 989.51 | 1.23 | 988.28 | 1.21 | 987.07 | 1.17 | 985.90 | 1.13 | 984.77 | 1.11 |
|     | 0.14   |      | 0.14   |      | 0.13   |      | 0.13   |      | 0.14   |      | 0.14   |      | 0.15   |      | 0.16   |      | 0.16   |      | 0.17   |      | 0.18   |      | 0.18   |      |
| 15° | 999.05 | 1.51 | 997.54 | 1.46 | 996.08 | 1.42 | 994.66 | 1.38 | 993.28 | 1.35 | 991.93 | 1.30 | 990.63 | 1.27 | 989.36 | 1.24 | 988.12 | 1.21 | 986.91 | 1.18 | 985.73 | 1.14 | 984.59 | 1.12 |
| 16° | 0.14   |      | 0.14   |      | 0.15   |      | 0.15   |      | 0.15   |      | 0.16   |      | 0.16   |      | 0.17   |      | 0.17   |      | 0.18   |      | 0.19   |      | 0.19   |      |
|     | 998.90 | 1.50 | 997.40 | 1.46 | 995.94 | 1.43 | 994.51 | 1.38 | 993.13 | 1.35 | 991.78 | 1.31 | 990.47 | 1.27 | 989.20 | 1.25 | 987.95 | 1.21 | 986.74 | 1.19 | 985.55 | 1.15 | 984.40 | 1.13 |
| 17° | 0.16   |      | 0.16   |      | 0.16   |      | 0.16   |      | 0.17   |      | 0.17   |      | 0.18   |      | 0.18   |      | 0.19   |      | 0.19   |      | 0.20   |      | 0.20   |      |
|     | 998.74 | 1.50 | 997.24 | 1.46 | 995.78 | 1.43 | 994.35 | 1.38 | 992.97 | 1.36 | 991.61 | 1.31 | 990.30 | 1.28 | 989.02 | 1.25 | 987.17 | 1.22 | 986.55 | 1.19 | 985.36 | 1.16 | 984.20 | 1.14 |
| 18° | 0.17   |      | 0.17   |      | 0.16   |      | 0.17   |      | 0.17   |      | 0.18   |      | 0.18   |      | 0.19   |      | 0.20   |      | 0.21   |      | 0.22   |      | 0.22   |      |
|     | 998.57 | 1.50 | 997.07 | 1.46 | 995.61 | 1.42 | 994.19 | 1.39 | 992.80 | 1.36 | 991.44 | 1.32 | 990.12 | 1.28 | 988.84 | 1.26 | 987.58 | 1.23 | 986.35 | 1.20 | 985.15 | 1.17 | 983.98 | 1.14 |
| 19° | 0.18   |      | 0.18   |      | 0.19   |      | 0.19   |      | 0.19   |      | 0.19   |      | 0.20   |      | 0.20   |      | 0.20   |      | 0.21   |      | 0.21   |      | 0.22   |      |
|     | 998.39 | 1.50 | 996.89 | 1.46 | 995.43 | 1.43 | 994.00 | 1.39 | 992.61 | 1.36 | 991.25 | 1.32 | 989.93 | 1.29 | 988.64 | 1.26 | 987.38 | 1.23 | 986.15 | 1.21 | 984.94 | 1.18 | 983.76 | 1.16 |
| 20° | 0.19   |      | 0.19   |      | 0.19   |      | 0.19   |      | 0.19   |      | 0.20   |      | 0.20   |      | 0.21   |      | 0.22   |      | 0.23   |      | 0.24   |      | 0.24   |      |
|     | 998.20 | 1.50 | 996.70 | 1.46 | 995.24 | 1.43 | 993.81 | 1.39 | 992.42 | 1.36 | 991.06 | 1.33 | 989.73 | 1.29 | 988.44 | 1.27 | 987.17 | 1.24 | 985.93 | 1.22 | 984.71 | 1.19 | 983.52 | 1.16 |

TABLE I (continued) International alcoholic strength at 20°C  
Table of apparent densities of ethanol-water mixtures - Pyrex pycnometer Densities at t°C, corrected for air buoyancy

| t°  | Alcohol % by volume |      |        |      |        |      |        |      |        |      |        |      |        |      |        |      |        |      |        |      |        |      |        |      |  |  |   |  |  |    |  |  |    |  |
|-----|---------------------|------|--------|------|--------|------|--------|------|--------|------|--------|------|--------|------|--------|------|--------|------|--------|------|--------|------|--------|------|--|--|---|--|--|----|--|--|----|--|
|     | 0                   |      | 1      |      |        | 2    |        |      | 3      |      |        | 4    |        |      | 5      |      |        | 6    |        |      | 7      |      |        | 8    |  |  | 9 |  |  | 10 |  |  | 11 |  |
| 20° | 998.20              | 1.50 | 996.70 | 1.46 | 995.24 | 1.43 | 993.81 | 1.39 | 992.42 | 1.36 | 991.06 | 1.33 | 989.73 | 1.29 | 988.44 | 1.27 | 987.17 | 1.24 | 985.93 | 1.22 | 984.71 | 1.19 | 983.52 | 1.16 |  |  |   |  |  |    |  |  |    |  |
| 21° | 0.20                |      | 0.20   |      | 0.20   |      | 0.20   |      | 0.21   |      | 0.21   |      | 0.21   |      | 0.22   |      | 0.22   |      | 0.23   |      | 0.24   |      | 0.24   |      |  |  |   |  |  |    |  |  |    |  |
|     | 998.00              | 1.50 | 996.50 | 1.46 | 995.04 | 1.43 | 993.61 | 1.40 | 992.21 | 1.36 | 990.85 | 1.33 | 989.52 | 1.30 | 988.22 | 1.27 | 986.95 | 1.25 | 985.70 | 1.23 | 984.47 | 1.19 | 983.28 | 1.18 |  |  |   |  |  |    |  |  |    |  |
| 22° | 0.21                |      | 0.21   |      | 0.21   |      | 0.21   |      | 0.21   |      | 0.22   |      | 0.22   |      | 0.23   |      | 0.24   |      | 0.24   |      | 0.24   |      | 0.26   |      |  |  |   |  |  |    |  |  |    |  |
|     | 997.79              | 1.50 | 996.29 | 1.46 | 994.83 | 1.43 | 993.40 | 1.40 | 992.00 | 1.37 | 990.63 | 1.33 | 989.30 | 1.31 | 987.99 | 1.28 | 986.71 | 1.25 | 985.46 | 1.23 | 984.23 | 1.21 | 983.02 | 1.18 |  |  |   |  |  |    |  |  |    |  |
| 23° | 0.22                |      | 0.22   |      | 0.23   |      | 0.23   |      | 0.23   |      | 0.23   |      | 0.24   |      | 0.24   |      | 0.24   |      | 0.25   |      | 0.26   |      | 0.25   |      |  |  |   |  |  |    |  |  |    |  |
|     | 997.57              | 1.50 | 996.07 | 1.47 | 994.60 | 1.43 | 993.17 | 1.40 | 991.77 | 1.37 | 990.40 | 1.34 | 989.06 | 1.31 | 987.75 | 1.28 | 986.47 | 1.26 | 985.21 | 1.24 | 983.97 | 1.20 | 982.77 | 1.20 |  |  |   |  |  |    |  |  |    |  |
| 24° | 0.24                |      | 0.23   |      | 0.23   |      | 0.23   |      | 0.24   |      | 0.24   |      | 0.24   |      | 0.25   |      | 0.26   |      | 0.26   |      | 0.27   |      | 0.29   |      |  |  |   |  |  |    |  |  |    |  |
|     | 997.33              | 1.49 | 995.94 | 1.47 | 994.37 | 1.43 | 992.94 | 1.41 | 991.53 | 1.37 | 990.16 | 1.34 | 988.82 | 1.32 | 987.50 | 1.29 | 986.21 | 1.26 | 984.95 | 1.25 | 983.70 | 1.22 | 982.48 | 1.20 |  |  |   |  |  |    |  |  |    |  |
| 25° | 0.24                |      | 0.25   |      | 0.24   |      | 0.25   |      | 0.24   |      | 0.25   |      | 0.26   |      | 0.26   |      | 0.26   |      | 0.27   |      | 0.28   |      | 0.28   |      |  |  |   |  |  |    |  |  |    |  |
|     | 997.09              | 1.50 | 995.59 | 1.46 | 994.13 | 1.44 | 992.69 | 1.40 | 991.29 | 1.38 | 989.91 | 1.35 | 988.56 | 1.32 | 987.24 | 1.29 | 985.95 | 1.27 | 984.68 | 1.26 | 983.42 | 1.22 | 982.20 | 1.21 |  |  |   |  |  |    |  |  |    |  |
| 26° | 0.25                |      | 0.25   |      | 0.26   |      | 0.25   |      | 0.26   |      | 0.26   |      | 0.26   |      | 0.26   |      | 0.28   |      | 0.28   |      | 0.28   |      | 0.30   |      |  |  |   |  |  |    |  |  |    |  |
|     | 996.84              | 1.50 | 995.34 | 1.47 | 993.87 | 1.43 | 992.44 | 1.41 | 991.03 | 1.38 | 989.65 | 1.35 | 988.30 | 1.32 | 986.98 | 1.31 | 985.67 | 1.27 | 984.40 | 1.26 | 983.14 | 1.24 | 981.90 | 1.22 |  |  |   |  |  |    |  |  |    |  |
| 27° | 0.26                |      | 0.26   |      | 0.26   |      | 0.27   |      | 0.27   |      | 0.27   |      | 0.27   |      | 0.28   |      | 0.28   |      | 0.29   |      | 0.30   |      | 0.30   |      |  |  |   |  |  |    |  |  |    |  |
|     | 996.62              | 1.50 | 995.12 | 1.47 | 993.65 | 1.43 | 992.22 | 1.41 | 990.81 | 1.38 | 989.40 | 1.35 | 988.00 | 1.32 | 986.60 | 1.31 | 985.20 | 1.27 | 983.80 | 1.26 | 982.40 | 1.24 | 981.00 | 1.22 |  |  |   |  |  |    |  |  |    |  |